

JOSHUA KIRBY

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[LinkedIn](#) • [GitHub](#) • [Portfolio](#)

Bilingual (English/Japanese) BSc Artificial Intelligence student on track for First Class.
Experienced in deploying production RAG architectures and full-stack AI applications.

EDUCATION

University of Birmingham
BSc Artificial Intelligence & Computer Science

Expected 2027
First Class Level (85% 2nd Year Average)

TECHNICAL SKILLS

- **Languages:** TypeScript, Python, SQL, JavaScript
- **Core Stack:** Next.js, React, Node.js, Tailwind CSS, Drizzle ORM
- **AI Engineering:** RAG Architectures, OpenAI API / SDKs, Vercel AI SDK, Vector DBs (Pinecone)
- **Cloud & Infra:** AWS (Lambda, DynamoDB, S3, EC2), Cloudflare Workers, SST, Serverless Architecture
- **Tools:** Git, Docker, PostHog, Vitest

EXPERIENCE

Genki Japanese and Culture School | [View Case Study](#)
Software Engineer

Fukuoka, Japan
Sep 2023 – Jun 2024

- Reduced staff burden by independently designing and deploying a production RAG chatbot across 2 Japanese schools, handling **14,000+ messages** from **2,000+ users**.
- Saved **~70 hours/week** by automating responses to repetitive queries, allowing staff to focus on high-value sales.
- Future-proofed the chatbot with an autonomous scraping pipeline that updates the database as website content evolves, requiring zero maintenance.
- Increased retrieval precision from **43%** to **62%** by conducting an ablation study (**700+** judgments) to validate keyword injection and query expansion techniques.
- Implemented safety guardrails that identify the AI's limitations on critical questions and transfers full context to staff, ensuring accuracy without asking users to repeat themselves.
- Tech: *Next.js, AWS, TypeScript, Python.*

The HALO Trust | Humanitarian Demining
Technology Intern

Dumfries, UK
Dec 2021 – Apr 2023

- Built training datasets for a **geospatial demining AI**, generating **1,500+ labels** on satellite imagery to calibrate **Mask R-CNN** models for structure detection in Afghanistan.
- Produced “Digital Twins” of conflict zones to model explosive risks for **12,000+ civilians**, directly supporting successful **UN humanitarian funding** proposals.

PROJECTS

AI Conversational Placement Test | [View Case Study](#)
Solo Project

Pilot
2024 – Present

- Overcame domain expertise gaps by interviewing a senior teacher to reverse-engineer their methodology, translating the “Ask → Hint → Explicit” pattern into a deterministic state machine.
- Slashed the student placement lifecycle by **99%** (2 weeks → 30 mins) by replacing manual Zoom interviews with an on-demand AI assessment.
- Scaled the platform for multi-school usage by designing a **multi-tenant** system with configurable authentication, ensuring data governance across distinct organisations.
- Achieved **70% strict recall** on grammar pattern extraction by building a custom eval suite (~**600** annotated samples) and a multi-stage “generate-then-validate” pipeline.
- **Guaranteed stability** by developing a versioning system for test configurations, allowing atomic deployments and rollbacks.
- Tech: *Next.js, Cloudflare, AWS, GCP Vertex AI, TypeScript, Gemini.*

ADDITIONAL INFORMATION

- **Languages:** Native English and Japanese speaker
- **Driving Licenses:** Full UK and Japan licenses